(1) Find the center and radius of the circle $x^2+y^2-x+2y-3=0$

Solution : We know that the center of the circle $x^2+y^2+2gx+2fy+c=0$ is $(-g,-f)$ and its radius is $r=\sqrt{g^2+f^2-c}$

Comparing given equation with $x^2+y^2+2gx+2fy+c=0$ we get,

$g=-\frac{1}{2}, f=1, \quad c=-3$

$\therefore$ center is $\left(\frac{1}{2},-1\right)$ and radius $r=\sqrt{\left(-\frac{1}{2}\right)^2+(1)^2-(-3)}=\sqrt{\frac{1}{4}+1+3}=\frac{\sqrt{17}}{2}$ unit.

2) Find the equation of circle passing through the point of intersection of the lines $x+3y=0$ and $2x-7y=0$ and whose centre is the point of intersection of the lines $x+y+1=0$ and $x-2y+4=0$.

Solution : The point of intersection of $x+3y=0$ and $2x-7y=0$ is (0, 0) { How ?}

Solving $x+y+1=0$ and $x-2y+4=0$ we get , (-2, 1) as their point of intersection.

The center of the circle is (-2,1) and it passes through (0,0)

$\therefore$ Its radius is

$\sqrt{(-2-0)^2+(1-0)^2} = \sqrt{4+1} = \sqrt{5}$

The equation of the circle is (centre-radius form)

Centre $(-2,1) \equiv (h,k)$, radius $r=\sqrt{5}$

$(x-h)^2+(y-k)^2=r^2$

$(x+2)^2+(y-1)^2=(\sqrt{5})^2$

$x^2+y^2+4x-2y=0$

3) Find the equation of circle, the end points of whose diameter are the centers of circles $x^2+y^2+6x-14y-1=0$ and $x^2+y^2-4x+10y-2=0$.

Solution : The center of the circle $x^2+y^2+6x-14y-1=0$ is $(-g,-f) \equiv$

The center of the circle $x^2+y^2-4x+10y-2=0$ is $(-g,-f) \equiv (2,-5)$

Scanned with CamScanner

The end points of the diameter of the required circle are $(-3,7)$ and $(2,-5)$ ..... The equation of the required circle is

$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$ (diameter form)

$(x+3)(x-2)+(y-7)(y+5)=0$

$x^2-2x+3x-6+y^2+5y-7y-35=0$

$x^2+y^2+x-2y-41=0$

4) Find the equation of the circle passing through points $(5,7),(6,6)\&(2,-2)$.

Solution : Let the equation of circle be $x^2+y^2+2gx+2fy+c=0$

This passes through the point $(5,7),(6,6)(2,-2)$ — (1)

Substituting co-ordinates of the point in the equ^n (1)

$25+49+10g+14f+c=0$

$74+10g+14f+c=0$ —— (2)

$36+36+12g+12f+c=0$

$72+12g+12f+c=0$ —— (3)

$4+4+4g-4f+c=0$

$8+4g-4f+c=0$ —— (4)

equ^n (2) − equ^n (3)

$2-2g+2f=0$ —— (5)

equ^n (3) − equ^n (4)

$64+8g+16f=0$ —— (6)

Solving (5) and (6)

$f=-3,\ g=-2$

Put $f=-3,\ g=-2$ in equ^n (4)

$c=-12$

Substituting the values of g, f, c in (1)

$x^2+y^2+2(-2)x+2(-3)y-12=0$

$x^2+y^2-4x-6y-12=0$

It is the required equation of circle.

5) Find the co-ordinates of the focus, equation of the directrix, length of Latus Rectum, and the co-ordinates of the end points of the Latus Rectum of the parabola $i)5y^2=24x,\quad ii)x^2=12y,\quad iii)3y^2=-16x$.

Scanned with CamScanner

Solution : i) $5y^2 = 24x$

Comparing $y^2 = \frac{24}{5}x$ with $y^2 = 4ax$, we get $4a = \frac{24}{5}$ $\therefore a = \frac{6}{5}$

The co-ordinates of the focus are $(a,0) \equiv \left(\frac{6}{5},0\right)$,

Equation of the directrix is $x = -a$, $\therefore$ $x = -\frac{6}{5}$ $\therefore$ $5x+6=0$

The length of Latus Rectum is $4a = \frac{24}{5}$

The co-ordinates of the end points of the Latus Rectum are $(a,2a)$ and $(a,-2a)$, $\left(\frac{6}{5},\frac{12}{5}\right)$ and $\left(\frac{6}{5},-\frac{12}{5}\right)$

ii) $x^2 = 12y$

The equation of parabola is of the form $x^2 = 4by$

$\therefore 4b = 12$

$\therefore b = 3$

Focus is $(0,b) \equiv (0,3)$

Equation of directrix is

$y + b = 0$ ie. $y+3=0$

l(Latus Rectum) $= 4b = 12$

End points of Latus Rectum are

$L(2b,b) \equiv (6,3)$

$L'(-2b,b) \equiv (-6,3)$

iii) $3y^2 = 16x$

$y^2 = \frac{16}{3}x$

The equ^n of form $y^2 = 4ax$

$\therefore 4a = \frac{16}{3}$

$\therefore a = \frac{16}{4\times 3}$

$\therefore a = \frac{4}{3}$

Focus is $(a,0) \equiv \left(\frac{4}{3},0\right)$

Equation of directrix

$x - a = 0$ i.e., $x - \frac{4}{3} = 0$

ie, $3x - 4 = 0$

l(Latus Rectum) $= 4a = 4\times\frac{4}{3}$

$= \frac{16}{3}$

End points of Latus Rectum are

$L(a,2a) \equiv \left(\frac{4}{3},\frac{8}{3}\right)$

$L'(a,-2a) \equiv \left(\frac{4}{3},-\frac{8}{3}\right)$

6) Find the co-ordinates of the focus, equation of the directrix, length of Latus Rectum, & the co-ordinates of the end points of the Latus Rectum of the parabola $x^2+4x+4y+16=0$.

Scanned with CamScanner

Solution : Rewrite equation $x^2+4x+4y+16=0$ as $x^2+4x=-4y-16$

$\therefore\ x^2+4x+4=-4y-12$

$\therefore\ (x+2)^2=-4(y+3)$

Now shifting origin to the point $(-2,-3)=(h,k)$ we get $x+2=X$ and $y+3=Y$ —(1) where $x=X+h$, $y=Y+k$

Equation of parabola become $X^2=-4Y$ which is of the form $X^2=-4bY$, $4b=4 \Rightarrow b=1$

Co-ordinates w.r.t new co-ordi system are focus $(0,-b)=(0,-1)=(X,Y)$ But $x+2=X$, $y+3=Y$

$x+2=0$, $y+3=-1$

$\therefore x=-2$, $y=-4$

Focus $(-2,-4)$

Equ^n of directrix is $Y-b=0$ i.e $y+3-1=0$ ie $\boxed{y+2=0}$

l (latus Rectum) $=4b=4$

End Points of latus Rectum

$L(2b,-b)=(2,-1)=(X,Y)$

But $x+2=X$, $x+2=2$, $x=0$

$Y+3=Y$, $Y+3=-1$, $y=-4$

$L(0,4)$

$L'(-2b,-b)=(-2,-1)=(X,Y)$

$x+2=X$, $x+2=2$, $x=-4$

$y+3=Y$, $y+3=1$, $y=-4$

$L'(-4,-4)$

7) Find the area of triangle formed by the lines joining the vertex of the parabola $x^2 = 12y$ to the ends of its Latus rectum.

Solution :

Given Parabola is

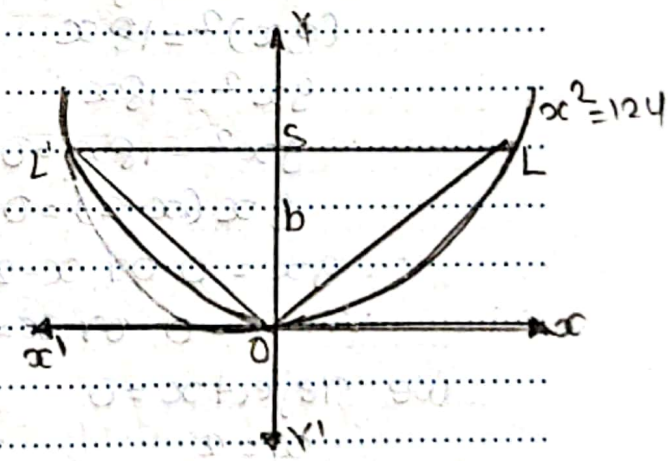
$x^2=12Y$

It is of the form

$x^2=4bY$

$\therefore 4b=12$

$\therefore \boxed{b=3}$



$LL'=l$ (latus Rectum)

$LL'=4b=12$

Also $OS=b=3$

$A(\Delta OLL')=\frac{1}{2}\times$ base $\times$ height

$=\frac{1}{2}\times LL'\times OS$

$=\frac{1}{2}\times 12\times 3=18$ sq. units

Scanned with CamScanner

8) For which point of the parabola $y^2 = 18x$ is the ordinate equal to 3 time the abscissa?

Solution : Given equation of parabola is

$Y^2 = 18x$ - (1)

We know that abscissa = x Co-ordinate of a Point and ordinate = Y co-ordinate of a point.

we have to find the co-ordinate of a Point on $y^2 = 18x$ such that $y = 3x$

$\therefore$ Substituting $Y = 3x$ in (1)

$(3x)^2 = 18x$

$9x^2 = 18x$

$9x^2 - 18x = 0$

$9x(x-2) = 0$

$\therefore 9x = 0$ or $x - 2 = 0$

$\therefore x = 0$ or $x = 2$

We reject $x = 0$

$\therefore x = 2$

$\therefore y = 3x = 3(2) = 6$

$\therefore$ The required point is $(2, 6)$

Sign of Teacher :

Scanned with CamScanner